

Theoretical Foundations for Website Design Courses

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Theoretical foundations in website design courses can facilitate students learning the genres of Internet communication. Genre theory and activity theory provide opportunities for emphasizing the social context of websites in ways that emphasize for students the identity-laden nature of sites that can often appear to be anonymous and addressed to multiple audiences. After proposing ways that these theories can be integrated into website design courses, the article focuses on two students' website portfolios and ways they utilize theories discussed in class to produce websites in two different fields.

Introduction and Background

A student took my course "The Rhetoric of Internet Publishing" in the fall of 1999. Steve had already had some experience with designing sites but was very interested in learning about online sales strategies. His ultimate goal was to make a living selling some type of product online. While my course did not cover JavaScript, he developed an interest in sites that used the "shopping cart" method for buying products, and he developed a simple site to sell ink jet printer cleaning supplies. The site design was a one-screen, table-formatted, three-color style with simple, clear language. After working with sites offering credit card services, he arranged for customers to purchase the product with credit cards and promoted his site as a fully operational online store.

Almost immediately, with the help of word-of-mouth marketing strategies, Steve's site was a success. The online business grew so rapidly that Steve quickly had to look for ways to invest his newfound capital. He had to incorporate his business and find someone else to

assist him with all the orders pouring in. The new life Steve is experiencing now is the result of determination and skill as a business manager on his own part, but it is also the result of effective communication and marketing via the Internet.

I can't take any credit for Steve's success, but in our class, we all discussed the importance of learning website genres and using that knowledge to create sites that would appeal to audience need. The process of analyzing an Internet audience is a difficult one, and it's even more difficult to design sites that appeal to the desired Internet community. But doing so gives the writer an enormous amount of power when done successfully. Steve is one example of how powerful one can be (in a culture that privileges economic success and power) when an Internet audience is addressed effectively. Just as with other genres, students have to be taught to understand and implement the genres of power regarding Internet communication, then to critique those genres.

In this article, I argue that two theories—genre theory and activity theory—provide opportunities for emphasizing the social context of websites in ways that emphasize for students the identity-laden nature of sites that can often appear to be anonymous and addressed to multiple, sometimes non-identifiable audiences. Genre theory provides benefits that rhetorical theory in general does not emphasize; genre theory, particularly the theories of the Australian genre school (Martin, Rothery, and Christie, for example), emphasizes the demystification of discourse conventions so that communicators can have broader access to genres of power. I stress here that a necessary convention for students to understand and apply in website design is demonstrating knowledge of social context in order to construct a site that reveals a complex awareness of audience. While rhetorical theory focuses on audience, purpose, and, in part, the complexities of communicating using the Aristotelian elements of *ethos*, *pathos*, and *logos*, the intense focus on social context and the demystification of discourse that genre theory and activity theory promote make them particularly useful theories in teaching website design. In addition, the rich analysis of the whole environment that activity theory fosters allows genres to be seen as tools that mediate activity, which is a more well-rounded focus than only rhetorical theory might contribute.

Theoretical foundations in website design courses can facilitate students' learning the genres of Internet communication; the theory provides context and justification, for example, when students consider what features to include in the construction of their sites. In addition, website design courses require inclusion of theory so that the courses are not only skills based but also grounded in theories that are important to the field of technical communication. Such a theoretical foundation helps ensure that students understand the reasoning behind practices of effective Web design, and a theoretical focus helps the teacher avoid the association with "skills-based" courses, an

association that has caused composition teachers, for example, to be devalued and feminized in the academy as workers charged with "cleaning up" the writing of students (see Bullock and Trimbur, *The Politics of Writing Instruction*).

Genre theory and activity theory both stress heavily the importance of social context. In their essay "Moving beyond the Academic Community," Chris Anson and Lee Forsberg study the adjustments writers make as they move from writing in academic settings to writing in professional environments. They stress the importance of social context awareness during this process, an awareness that preceded later work on this concept:

While certain surface-level writing skills are 'portable' across diverse contexts, such skills are less important to making a successful transition as a writer than coping with the unfamiliar epistemological, social, and organizational characteristics of a new context. A writer in such a context is in many ways 'illiterate' until he or she begins to understand these characteristics and their manifestation in written texts. Our research shows that becoming a successful writer is much more a matter of developing strategies for social and intellectual adaptations to different professional communities (201)

Studying the development of these strategies proved valuable for Anson and Forsberg as they discussed transitions writers make to communicating in new environments. Similar studies (such as Winsor's *Writing Like an Engineer*) have discussed ways that genre theory and analysis of social context have helped students adjust to new discourse communities. Similar strategies of analyzing and adapting to particular social contexts are essential for students moving from addressing an academic audience to addressing a more professional audience through the Internet.

Of course, the Internet is the perfect illustration of the need to adapt socially and intellectually to different communities because of the multitude of communities that exist; Anson and Forsberg's work highlights the importance of social context as writers adjust to writing for professional communities. Learning more about the audiences' social contexts enables the students to address those professional communities more effectively. (See Preece's work also for a discussion of social context in the development of online communities.) As Anson and Forsberg point out, this type of task results from students' awareness that writing is socially constituted (202).

What follows, then, is a proposal for integrating genre theory and activity theory into website design courses, with the focus being on encouraging students to analyze social context in ways that directly inform the design of their sites. Genres are addressed first, followed by situating genres as tools within activity systems in order to emphasize the important role genres as tools play in students' participation

within the activity system of a particular discourse community. While it is not possible to elaborate on both theories in great detail, the discussion highlights important parts of the theories to establish ways they might enhance instruction in website design courses. The goal of analyzing the genres is to help determine what activity is being mediated by the generic tools; then, the students will be able to see more clearly what common purposes are being accomplished within the activity systems as a result of the generic tools. Because this discussion focuses on genres as tools within the context of activity systems the students are addressing through their sites, I would like to situate the concept of genres within the context of activity theory by first defining genres and clarifying them.

Foundational Theories for Website Design Courses

Genre Theory

When I refer to “genre” here, I mean the following concepts:

Meeting Audiences’ Expectations

Website audiences have expectations regarding the ways websites in various fields are designed. For example, websites selling books have a search tool that allows users to search based on author, title, subject, or key words. Often, pictures of books are presented with the search results, along with price and availability. After visiting sites with these features and using them, audiences come to expect them to be present in general on sites that sell books. The search tool feature and pictures could be said to be generic characteristics of a genre in this field. These characteristics reflect needs that visitors to the site have; the genre is a result of the social context surrounding the need to buy books online.

Addressing Social Context

In order for genres to be successful, they need to address the social context of their audiences (Winsor; Berkenkotter and Huckin; Martin). Even though the audience will be very broad and public, rather than a narrowly defined discourse community, website designers can be aware of audience need and ways the audiences might use information gathered from the website. Such knowledge of social context, which genre theory privileges and emphasizes, contributes to the kinds of features used to construct certain website genres. A heightened awareness of audience and the social context(s) it comes from, then, is an important focus of genre.

Analyzing Discipline-Specific Discourse Communities

Genre theory promotes analyzing discipline-specific discourse and the community it represents (Berkenkotter and Huckin; Miller; Winsor). Therefore, one way students can learn about the social context of the audience for whom they will be writing is to analyze sites that already exist and identify characteristics and purposes of existing sites. Such analysis will prepare students to develop sites that better identify with the social context of their audiences as they interact with the sites.

Addressing the Relationship between Form and Content

An essential relationship exists between genres' form and content (Berkenkotter and Huckin). For example, links on websites might be considered part of the form of the page; however, they also help construct the content, because they assist in organizing the website in a way that helps the audience find information using hypertext, rather than linear, reading strategies.

Explicit Teaching/Learning and Technologically Advanced Genres

Constant learning and analysis have to occur when studying technologically advanced genres (as opposed to "natural literacy") (Cope and Kalantzis). This constant learning is necessary because of the inevitable evolution process that allows technological genres to evolve rapidly to meet audiences' changing needs. David Russell writes, "newcomers to a genre/activity bring their cultural history to their writing, and take an active role in learning as they wrestle with new genres" ("Writing and Genre" 229); the process of learning genres is a struggle, one in which teachers of writing can assist through explicit teaching and acknowledgment of genres' evolution. Howard Rheingold's explanations for technological advance—enabling technologies and convergence—underscore the potential for computer technology to change rapidly, providing an environment for genres to change as well in response to technological evolution. Rheingold states, "An *enabling technology* is one that makes another technology possible. Some powerful technologies spring into existence when they do because some kind of price or performance threshold has been crossed by one or several enabling technologies" (61). In addition, "convergence" occurs as a result of ideas intersecting and technological evolution (61). While Rheingold's ideas focus on the evolution of technology itself, they emphasize the potential for genres to change in response to the technology.

Identifying Cognitive Frameworks and Participating in Social Action

Genre theory stresses situated cognition and the need to identify with the cognitive frameworks that experienced audience members

(and users of genres) already have (Berkenkotter and Huckin; Winsor). The learner of genres is an actor participating in social action (Winsor; Miller; Berkenkotter and Huckin), an actor who must learn by not only analyzing a particular discourse community or audience but also participating within it.

Recognizing Genres as Tools within Activity Systems

As Russell points out, when theories of activity systems and genre are synthesized, "it is possible to see discourse (vocalizations and inscriptions) as one kind of tool among many others and to relate genres to other kinds of material actions" ("Rethinking" 513). Therefore, the writing and implementation of graphics used to design websites might be seen as tools used within an activity system. In addition, Russell writes that

[a] . . . step in connecting genre to activity theory—and expanding dialogic theory—is to see that written genres help mediate the actions of individuals with others in collectives (activity systems) to create stabilized-for-now structures of action and identity. ("Rethinking" 514)

Genres, therefore, might be seen also as tools for mediation.

Using genres within activity systems allows communicators to gain agency within disciplinary contexts. Russell writes,

Agency is distributed in streams of activity as participants appropriate voices in the networks of disciplinary practice. Their images of authorship change as they negotiate authorship among themselves in their oral and written interactions, redrawing disciplinary boundaries as they redraw their personal boundaries and align themselves with—and sometimes reject—powerful disciplinary social practices. ("Writing and Genre" 230)

Even though students in website design classes are often students acting within an academic context, teachers sometimes expect them to go outside the academic community and interact with professional activity systems at some point during the course, such as designing a site for professional purposes. The website design class, therefore, becomes a "contact zone" (Pratt) because the worlds of the academic and professional are intersecting. The instructional environment is academic, but the Internet community in which students are participating is professional.

Recognizing Typified Actions

When students, for example, recognize recurring actions and their purpose, they can understand the purposes behind recurring generic features. This concept links genre theory strongly with activity theory

because the activities themselves are brought into focus: the activities are what cause genres to change and what determine specific features that work within tools functioning within the system. As Russell writes, "Participants' shared recognition of the typified actions that a genre operationalizes is the key to distinguishing one genre from another" ("Rethinking" 518).

Social processes may be causing the unique and variable website genre to evolve over time with particular features, similar to what Karlyn Campbell and Kathleen Jamieson addressed in some of the earliest work on speech genres; over time, genres evolve based on needs users of the genres have, as well as needs the creators of the genres have. The genres grow as the social situations that produce them change (Miller). While these scholars and others emphasize the evolution process that takes place with speech genres, this development also occurs with website construction; as technology continues to develop, and as Internet-savvy audiences' needs continue to change (such as with the development of search tools for the book-selling sites mentioned previously), website genres will continue to evolve to meet audiences' needs. The interrelationship between creator and user of website genres (and genres in general) causes genres to evolve dynamically to meet the needs of both parties simultaneously (ideally), although those needs might be different, such as in Web advertising, for example.

It is attention to specific features of Web design, as well as attention to the content of the page (Berkenkotter and Huckin emphasize the close connection between form and content), that will allow students to structure an effective website. As a result, students will be able to address specific features such as links and their organization, design of frames, inclusion of JavaScript, placement and labeling of graphics, as well as the rhetorical message that is being communicated. The relationship between the form and content elements is what defines the effectiveness of a particular website and its ability to identify itself with an audience.

Promoting the analysis of Internet genres in an historical/social way (an emphasis that Miller promotes regarding discourse communities) creates an environment for students to see the recurrent social needs that have caused a particular website genre to evolve. Awareness of these needs is important for students who are both learning about the development of website genres and attempting to contribute to that development, that activity system, by constructing websites. Learning what the audience's needs are allows students to participate in that community. This learning process requires analysis of audience, purpose, and the features of particular websites within a certain social context, such as sites selling a specific product or search engine websites. Learning HTML or website authoring programs simply is not enough for students to be able to address social context concerns adequately; students cannot be expected automatically to enter such a

virtual world of communication successfully. Such an idea supports the concept of "natural literacy" that genre theorists Bill Cope and Mary Kalantzis, J.R. Martin, Frances Christie, and Joan Rothery, for example, argue against through their work in systemic functional linguistics. Students have to be taught the genres that they learn, and since website genres and technology change so rapidly, students need to be given tools and principles to be able to analyze and produce genres over time.

Activity Theory

During the past few years, activity theory has become a way to explore the variety of social dynamics present as students and professional writers learn to write for different audiences (Spinuzzi; Russell, "Rethinking," "Writing and Genre," "Limits," "Looking Beyond," personal conversation). Activity theory is based on the idea of activity systems, that the whole action of writing and communicating is based on a system that affects and constructs the writing and communication. The social situation (the audience as well as the members of an academic or professional community that might influence the group or team that is communicating together), the social context (the occasion for writing that might be overlaid with history and meanings that need to be addressed in a particular document or communication medium [Chaiklin and Lave]), the tools used for writing (various writing genres and the technological tools used to produce them)—all of these things can make up activity systems (Winsor, "Genre and Activity Systems"; Cole and Engestrom; Engestrom; Engestrom et al.; Cole et al.). One valuable aspect of using activity theory to study the dynamics of website construction is its adaptability; it is not a "neat" theory that is fixed but allows for the change and "messiness" necessary for studying the social nature of writing and the social actors who work together to produce it.

Russell, in "Rethinking Genre in School and Society," defines an activity system as

any ongoing, object-directed, historically conditioned, dialectically structured, tool-mediated human interaction. Some examples are a family, a religious organization, an advocacy group, a political movement, a course of study, a school, a discipline, a research laboratory, and a profession. These activity systems are mutually (re)constructed by participants historically, using certain tools and not others, including discursive tools such as speech sounds and inscriptions. The activity system is the basic unit of analysis for both groups' and individuals' behavior, in that it analyzes the way concrete tools are used to mediate the motive (direction, trajectory) and the object (the *problem space* or the *focus*) of behavior and changes in it. (510)

Successful website designers have the potential to further the evolution of website genres (and thereby the activity systems that produce those genres) by introducing new and more effective features (see Cole and Engestrom for a discussion of the evolution of activity systems). New, effective features can serve almost to critique the past absence of those features for a particular audience and thus create new and more sophisticated website genres.

Activity systems include "mediating artifacts" (Engestrom), which might be seen as tools that help accomplish some sort of outcome. According to Russell, "Human consciousness—'mind'—grows out of people's joint activity with shared tools" ("Looking Beyond" 4). The use of the tools relates to shared values that community maintains as a culture; through using these tools, a group of people in an activity system share common motives for outcomes of the system (Russell, "Looking Beyond" 4–5). In this case, generic elements of websites are tools used to accomplish particular goals within activity systems, the goals being specific to the purposes of the activity systems. These goals may not be apparent at first but can be discovered more clearly upon analyzing the purposes of tools and motives within activity systems.

Russell states that

the tools that people in some activity system use together and the ways they use them change over time, historically, as they borrow new ways of working together from other activity systems or invent entirely new ways, transforming the activity potentially. ("Looking Beyond" 9)

Such a developing mode of instruction as website design benefits from an activity theory and genre analysis because students can begin to understand how evolving website genres can influence the developing activity systems, participating in various ways to accomplish some sort of outcome via the Internet. In addition, the ability to analyze and implement genres has the benefit of promoting the evolution of activity systems, and students can observe this process through application of these theories. Elsewhere, Russell emphasizes the critiquing process that can occur as students begin interacting socially with genres in activity systems ("Limits"). The critique is what leads to changes within the systems.

Activity theory allows a focus on broader goals that are being accomplished through communication. Students can explore why websites are constructed in certain ways, using generic tools, based on audience need. In addition, the powerful dynamic of technology and its influence on activity systems can be discussed, as well. As a result of technology, websites have developed new features, some of which may not be so beneficial but are there nevertheless because the technology is available. An example might be the existence of unnecessary graphics, due to the prevalence of free images online, as

well as graphics and animation programs. Should these unnecessary graphics help define website genres? Readily available technology presents difficulty in answering this question because ineffective communication strategies may yield undesirable website characteristics that may still be accepted (and expected) by members of the audience.

Sometimes, when only genres are explored, the characteristics of the genres are emphasized too much, rather than the problems, motives, and goals of the writers. Activity theory provides more of a context for studying the writing and its effects on audiences. According to Russell, the theory provides a texture, a weaving, a web of meaning (personal conversation). The genres themselves are tools that are considered to be part of various activities. Activity theory allows the genres under discussion to be viewed in a broader context; social context, then, can be addressed in a more fruitful way when it is seen in conjunction with a broader theory of systems, rather than just genres functioning within a particular context.

To illustrate the value of integrating genre theory and activity theory into website design courses, I will discuss two portfolios created by students in a website design class I teach at Tennessee Technological University (TTU), titled "The Rhetoric of Internet Publishing." The discussion of the portfolios focuses on how genre theory and activity theory apply to developing specific sites: over the course of the semester, both students developed an awareness of social context, based on theoretical principles we had discussed and applied in class. This awareness allowed the students to develop websites that addressed their intended audiences more effectively than if social context had not been emphasized in the course. In addition, they both attempted to participate in the activity system of Web page design for fields and audiences they chose to address, therefore contributing to the evolution and development of website genres in two fields.

Genre Theory, Activity Theory, and Their Inclusion in the Website Design Classroom

"The Rhetoric of Internet Publishing" provides students with opportunities to learn about specific audiences for websites on particular topics and then to design sites addressing those audiences. In order to accomplish this task, the students must acquire familiarity with the types of activity that the websites promote and ways the websites might be seen as tools within a particular activity. This type of knowledge can occur through analysis of the websites' audiences and the functions the websites seem to be serving within an activity system. In some ways, these analyses are speculative; it may not be possible to define with a large degree of certainty how features of a website genre correlate with the types of activity that site is promoting, but having students begin to notice the kinds of activity surround-

ing websites and then connect that activity in some ways with the features of the websites themselves allow students to have a better idea of how to construct their own sites that are based on similar activity purposes.

During the course, the students construct at least one website (with multiple pages) themselves. One goal of the course is for students to choose a particular Internet audience they would like to contribute to (or enter into dialogue with) and then create a website that addresses that audience. The students need to learn the importance of creating community, as well, which Jenny Preece discusses through her emphasis on online communities. The processes of analyzing the audience and then addressing it utilize principles of genre theory and document design that we have covered in class. For example, as the students learn about the audience, they must also learn about the social context of that audience (Martin; Berkenkotter and Huckin), as well as particular features that seem accepted by that audience to communicate the website's content (similar to the form and content relationship that Berkenkotter and Huckin discuss in their work). Some Internet sites, such as ones involved in sales, seem to use more "flashy" and animated graphics, while other sites, such as academic ones, tend not to. The students need to determine what kinds of features successfully communicate certain content and why, based on their analyses of the websites' audiences and the purposes of activity indicated by sites already published on the Web.

Kalina and Carolina are two students who designed pages using principles we discussed in class. Both students targeted specific audiences using genre principles designed to elicit audience response, particularly regarding social context. As a result, both students were able to participate in conversations involving Internet communities, and both students were able to see the application of theoretical principles to designing their sites. This process involved adjustment and learning on the part of the students; they had to learn the complexities of social context while at the same time respecting the dynamic nature of that context. As Anson and Forsberg discuss, "there is no such thing as a 'context' as a static entity, that is, a place or function whose rules are entirely visible and can be learned or known independently of the effect that the outsider creates by entering it" (225). The learning process also involves assessing the audience, as well as its activity, and analyzing it before entering it; the students helped accomplish this goal by studying websites in the fields they would be attempting to address. This process coordinates with the observations Anson and Forsberg made:

that the writer must first become a 'reader' of a context before he or she can be 'literate' within it. This literacy does not seem restricted to mutual knowledge of some intellectual domain (cf. Hirsch, 1987) but includes highly situational knowledge that can be gained only from participating in the context, which itself is in a constant state of change. (225)

Participating in this context also involves mediating among and between the cultural histories that the students bring to communication and the cultural histories that various audiences bring. Michael Cole and Yrjo Engestrom discuss cultural mediation by stating that

[it] has a recursive, bidirectional effect; mediated activity simultaneously modifies both the environment and the subject.

Cultural mediation implies a special importance of the social world in human development since only other human beings can create the special conditions needed for that development to occur.

A natural unit of analysis for the study of human behavior is activity systems, historically conditioned systems of relations among individuals and their proximal, culturally organized environments. (9)

Studying genres as tools can highlight purposes of activity systems and can provide a richer context for understanding audiences, Internet communities, and their purposes.

Throughout the analysis of sites below, some text features are mentioned; however, I want to clarify that according to Russell,

Texts recognized by participants in an activity system as belonging to some genre may not share a specific set of definable formal features (that is, what participants in that activity system, or others such as linguists, point to when asked how they know this is one kind of text and not another). Participants may recognize these texts as belonging to the same written genre as long as these texts are operationalizing the actions of participants in the activity system Participants' shared recognition of the typified actions that a genre operationalizes is the key to distinguishing one genre from another. ("Rethinking" 518)

Because the analysis that follows encompasses more than just features, the types of activity surrounding the social context become the most important concept hierarchically, with the textual and other format features falling underneath.

Carolina

Carolina designed a site that would appeal to readers interested in tennis in general and in TTU's tennis team specifically. The activity system in which she wished to participate might be people interested in knowing more about tennis at her institution. The purpose, then, was mainly informational, although Carolina decided to use her website for recruiting purposes as well. Carolina's site could be said to be similar in purpose to two other Tennessee state universities' tennis sites: The University of Tennessee (UT) and Middle Tennessee State University (MTSU). Briefly analyzing these sites reveals characteris-

tics that could be considered generic characteristics, although it is not possible to concretely define a genre based on such a limited analysis. Both of these sites present information to interested audiences to gain support of some kind for the women's tennis teams. The UT women's tennis site presents information about the team in the form of articles with short summaries with links to longer summaries, for example. The pictures, which perhaps were included to allow a more personal portrayal of the team, are in some ways impersonal, since the players are virtually anonymous. The summaries themselves pertain mostly to game results and winning strategies of the tennis team.

The UT women's tennis team site also contains a history page, which includes links to the roster, coach information, series records, and information on championships. The only recruitment information appears in a discussion of the "Lady Vol Boost-Her Club," which describes support for athletic scholarships as well as publicizing the team. The purpose of the site as a whole seems to be providing information on the tennis team. In contrast, the MTSU women's tennis team site provides a bit less information, but it does include a page on "Tennis History and Records," which includes the roster, 1999 results, 1999-2000 individual statistics, and 2000 results. Similar to the UT site, this site appears to focus primarily on the activity of disseminating information to people interested in the school's teams.

For sites like these that focus on the activity of information giving, some generic features might be links to articles, information on the history of the teams, and statistics. These sites give an indication of areas Carolina might address in her own site if she wanted to align her site with similar activities and purposes.

Carolina studied sites similar to these and others in order to identify the social context, purposes, and activities to which the audiences of the sites would be relating. However, Carolina seemed to take this knowledge and apply a more personal quality to her sites, one that reflected the importance of community within a team. Since Carolina is a team member, rather than a sportswriter or webmaster, she wanted to communicate the sense of community within the TTU tennis team, which she represented by using pictures taken during competitions. The pictures did not refer to specific team members' names, similar to the UT site, but the pictures helped to show familiarity with the larger community the team represents, that of the university as a whole. To emphasize this relationship, she referred to TTU's tennis history and its tennis schedule on her site. (See Figure 1.)

Carolina designed her site to attract women who might want to be a part of TTU's tennis team; in this sense, she took on an additional purpose compared to the UT and MTSU sites: she wanted to focus on recruitment instead of just information giving, although the UT site



Figure 1. Screen shot 1.

did include a page of information that contained details on support for team members. To persuade her audience through this information presenting/recruiting purpose, she did three things that show an awareness of social context: she used pictures that show happy events for the team, she emphasized the word "you" throughout her site (by telling the audience what will happen if they become part of the team), and she included a brief personal narrative about why she came to TTU and what benefits she had gained from her experiences on the team (see Figures 2 and 3).

Carolina's site illustrates how a genre might evolve and change, based on the purposes and goals of a particular activity system. While the other universities' sites were focused mainly on information giving, rather than recruiting, Carolina chose to address an audience who would be interested in the information and might be willing to join the team. The tone she incorporated (through use of the word "you" and her personal experience) helped to create a sense of community that might serve the purpose of her site: to inform as well as persuade others to join the team. Her website, therefore, contains different features and tone because it has a slightly different purpose (and consequently a different activity is being promoted), but she included features such as the tennis history and player information, like some of the other sites she studied. Because her site encompasses different

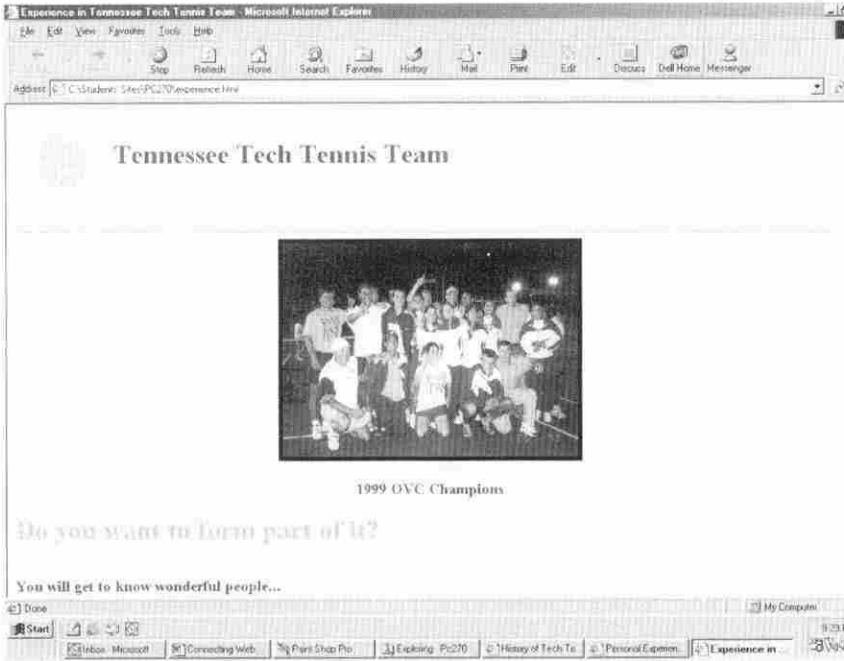


Figure 2. Screen shot 2.



Figure 3. Screen shot 3.

purposes and activities, it contains some different characteristics from the other websites she analyzed.

Kalina

Kalina was interested in designing a site that would help people who were visiting China to plan their trip. Before constructing her site, Kalina analyzed sites such as [chinatoday.com](#) and [internationalist.com](#). These sites were highly ranked when "traveling in China" was typed into the search engine [google.com](#).

These sites both exemplified characteristics that would be helpful for people planning trips to China. The activity system, then, might be potential travelers to China who would be using the sites to determine facts about the country as well as places to visit. The purpose of these sites is to convey information to people interested in performing similar activities (traveling to various places within China).

[Chinatoday.com](#) presents facts about China, such as populations for various cities. A table presents links grouped under the headings of "Provinces," "Autonomous Regions," "Municipalities," and "Special Administrative Regions." When the user clicks on one of these links, additional information about the particular area appears, along with a map that shows the location for the area. On the left side of the page is a frame with links users can implement to facilitate navigation of the site. One of those links is "Culture and Tradition," which includes links to "Customs," "Libraries, Museums, and Galleries," "Philately," "Related Links," "Organizations," and "Others." [Internationalist.com](#) provides information on many locations, but China has its own page on the site. The page presents information in table form with the headings of "Country Basics," "Key Cities," "Hotels," "Weather," "Books," "In-Country Search Engines," "Country Overview," "Maps," "Restaurants," "Airlines," and "Live Cams." The purpose for visiting this site is a bit different from the purpose of visiting [chinatoday.com](#), since [internationalist.com](#) does not emphasize as much information about the country itself, such as the population size. Provinces are emphasized in [chinatoday.com](#) whereas major cities are the focus of [internationalist.com](#). The activity systems addressed by these two sites, while not completely different, would have slightly different purposes in using the information.

Kalina began constructing her site from a different perspective because she is a Chinese-American who has invested time and effort in studying Chinese culture. When she designed her site, Kalina drew upon personal experience from visiting China as well as knowledge gained from being exposed to Chinese culture, as she was brought up in a Chinese-American home. Her site's contents reveal her attempts to identify with the "travel information" website genre in similar ways that the other sites mentioned did, by including descriptions, maps, and links of places her audience might like to see; links to sites her

tourist audience may be interested in; and advice to potential travelers. As Carolina did, Kalina first analyzed websites from travel agencies and organizations to understand the audience she would be addressing. This process is a sophisticated one, since the audience analysis is taking place indirectly: instead of interviewing the potential audience for her page (which ideally would be useful preparation for designing a site), she learned about the audience by studying common ways travel agencies address their audiences, particularly ones that address audiences traveling overseas to visit unfamiliar cultures. In this way, the audience she was attempting to address is very specific, not an overly general one, such as the audience for *travelocity.com*, which might include anyone interested in traveling anywhere. Kalina was specifically interested in learning about the social context of audiences who will be visiting unfamiliar cultures. She extrapolated from the form and content of travel information websites, like *chinatoday.com* and *internationalist.com*, to define her audience further, as can be seen by the various pages she created.

Kalina's "Tourist Attractions" page contains descriptions of four areas her audience might want to visit (see Figures 4 and 5). She included a list of cities for each area, places and items of interest for each city, and even a list of reasons why people should visit. Within the table are additional links for points of interest Kalina thought her audience might like to know about. The tabular format Kalina implemented for this information is especially effective because it allows the viewer to scan the information quickly. The table serves as a navigation tool for the audience, as it did on other sites she analyzed. (See Rheingold and Nielsen for a discussion of navigation and its development as readers explore virtual worlds.) The city names are the headings for each table column, with the places and items of interest following vertically underneath each heading.

The "Chinese Culture" page begins with an explanation of why a reader might want to read the traveling advice Kalina included (see Figures 6 and 7). She took this opportunity to clarify for the reader exactly why the information would be useful, since she is addressing an audience who does not have experience with Chinese culture. Under each major heading of "People," "Eating," and "Customs," she not only stated facts about the culture but also illustrated how visitors might use that information to make their visit to China more successful. This particular page correlates to the page on culture from *chinatoday.com*, although Kalina's page is much less detailed. Because Kalina knew her readers might like to know more information than her site can provide, she included links to *The Hong Kong Standard Newspaper*, the Hong Kong website, and the Hong Kong Tourism Association.

Through her portfolio, Kalina indicated a knowledge of her audience's needs: people visiting China for the first time would want to know points of interest and advice on adapting to the culture. In a



Figure 4. Screen shot 4.

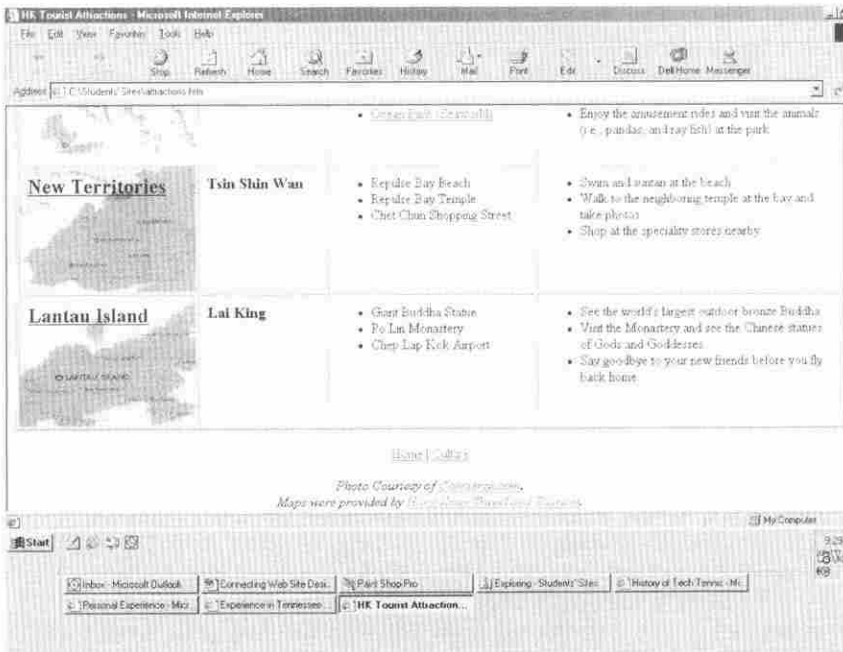


Figure 5. Screen shot 5.



Figure 6. Screen shot 6.

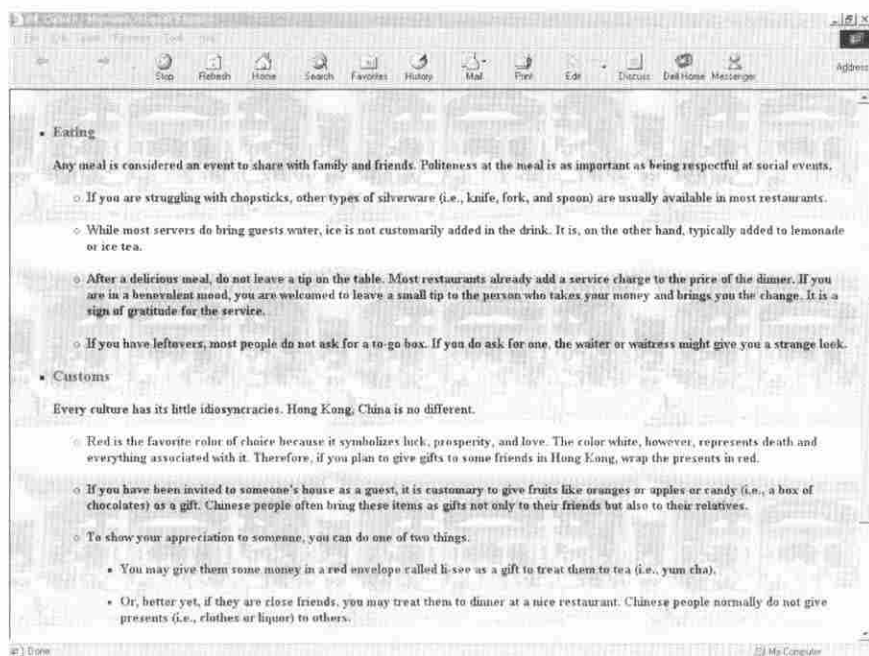


Figure 7. Screen shot 7.

complex way, she addressed the tourist audience by appealing to their needs through the "tourist" Web page genre, specifically adapted to an international context, but she also was communicating a knowledgeable representation of a culture with which she affiliates herself, so she is identifying with that social context as well. The cultural history that participants in activity systems bring to communicative contexts is a tenet of cultural-historical activity theory (Cole and Engestrom), and Kalina illustrates how her nationality can play a part in the cultural background she brings to the development of her site. Kalina negotiated the various aspects of social contexts through the form and content of her pages.

Awareness of generic website characteristics that would target their audiences, in addition to knowledge of document design, helped prepare these students for creating more effective websites. In addition, these students demonstrated skill in encouraging audience interaction with their sites: because the Internet can cause distance to exist between writer and user, interaction needs to be fostered to produce "responsive environments" (Rheingold 118). Both students attempted to create such environments through the use of links and, in Carolina's case, the "you" statements, which drew the audience in as participants.

One obvious follow-up strategy necessary for evaluating the effectiveness of these sites would be to have them published online and determine their usefulness in some way, whether from user response through commenting or responding to prompts (like Carolina's guest book, her request for scholarship information, or Kalina's request for e-mail comments). However, neither of these students published their sites (Kalina published hers briefly on the TTU server as part of an honors project). Teaching students to be aware of website genres and then replicate them does entail some sort of evaluation through publication, and while this important assessment did not occur with this class formally, it is an important step that should be included in future classes, I believe. Such an evaluation would help determine if audiences participating in that activity system were able to determine ways to participate effectively.

Conclusion

Teachers of website design and technical communication have the perfect opportunity to help students become social actors who are learning the social contexts of writing in the professions. We can help them because we are with them physically and interact with them in the classroom; such an approach is particularly useful in the Internet instruction classroom, where a teacher could directly assist students during their processes of creating websites and learning social contexts of communication. Helping students identify powerful genres of

communication and then construct them using technology can help us develop students who are better able to think critically and holistically about academic concepts. Such critique can enable newcomers to particular fields to revise activity systems and encourage them to develop and evolve. By being explicit with students about ways to address the concept of social context through website genres, for example, we can help students learn about social context and writing in ways that will help them not only in the classroom but also outside it as they work to apply concepts learned in academia to the workplace. In addition, we will be helping to ensure that disciplinary theoretical knowledge continues to form the foundation for pedagogical application.

There is a growing need for website design teachers who are sensitive to the needs of communicators in a rapidly changing, technologically advanced society. Rheingold writes,

communications and cultures, by their nature, are locked into a cycle of perpetual corevolution. Technologies that make it easier to create new technologies are self-accelerating. And the ripples of effects are not limited to technologies. Ways of life change, public perceptions change, the limits of what is possible or permitted change, when a new communication medium is adopted by a population. New technologies enable new cultural institutions as well as newer technologies The entire roller-coaster of twentieth-century history has been driven by the raw transformative power of innovation in communication technology. There is no reason to suspect that this continual and highly unpredictable process of technological and cultural change will cease during the next ten to twenty years. (218)

Rheingold published *Virtual Reality* in 1991. Approximately ten years later, his words have proven to be true. Technological advancements are still unpredictable, and as teachers of communication in a website design era, we have the responsibility to teach students the tools for navigating the creation of genres within virtual systems of power and activity.

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